# Auditable Empirical Evaluation of Machine Unlearning: Immutable Deletion Manifests, Exact-Retraining Ensembles, and a Reproducible Fashion-MNIST Pilot

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**Pilot and methodology preprint.** This manuscript reports retrospective P001 infrastructure and exploratory observations. It establishes neither successful nor certified unlearning.

**Abstract**

Machine-unlearning evaluations often compare an approximate intervention with a single retrained model, despite stochastic training producing a population of counterfactual models. This practice can obscure both natural training variability and a more basic identifiability problem: a deletion request may not measurably change the learned function under the selected assay. We present an evidence-traceable evaluation workflow that binds deletion requests, retained sets, seeds, configurations, checkpoints, predictions, metrics, and derived figures to hash-verified experiment records. The workflow represents exact retraining with an ensemble and separates reference-assay sensitivity from subsequent method evaluation.

We validate the workflow in P001, a retrospective pilot using a reduced Fashion-MNIST split, one frozen one-percent IID deletion manifest, a small convolutional network, one original training seed, and 15 independent exact-retraining seeds. The predefined $k = 10 \rightarrow 15$ seed-count diagnostic changed by 4.48% for mean Jensen–Shannon divergence and 3.14% for prediction disagreement, below the frozen 10% pilot threshold. Although three approximate methods were executed, retrospective analysis found that the deletion itself was not separable from exact-retraining variability under these two output metrics. Method proximity to the exact ensemble therefore cannot be interpreted as evidence of successful removal in this condition. High accuracy on IID-deleted examples is likewise not direct evidence of memorization because held-out examples may remain classifiable from shared class structure.

P001 validates infrastructure integrity and initial reference-ensemble stability, not certified unlearning, privacy equivalence, or generalization. A subsequent prospective calibration found that JS divergence could identify full deletion of two tested classes, but a candidate exact/exact-percentile equivalence margin failed to detect the frozen one-percent residual-class violation. The evidence therefore supports distinguishability analysis but not an equivalence claim. The resulting archive keeps the confirmatory method matrix blocked until its decision rule is defensible.

## 1 Introduction

Approximate machine unlearning is intended to remove selected training influence without the cost of complete retraining. The relevant counterfactual is a model trained on the retained data, but stochastic optimization produces a distribution of such models. This observation changes the evaluation problem. A single retrained network is one draw from a random training procedure, not a unique parameter-space target. Differences caused by initialization, minibatch order, optimization, and augmentation can be comparable to or larger than those caused by deleting a small set of observations.

Many empirical evaluations emphasize retained-set utility, decreased performance on a forget set, or membership-inference behavior. Each measurement can be useful, but none is identical to the counterfactual question: would the observable and latent behavior of the post-unlearning model be statistically consistent with models trained without the deleted records? A model can preserve test accuracy while changing its confidence distribution; an attack can approach chance while the model remains behaviorally unlike

retraining; and an IID-deleted example can remain correctly classified even when it was never used for training. These possibilities motivate a multidimensional target rather than a single forgetting score.

This study separates two questions that are often combined. First, is the deletion request itself empirically identifiable under the selected model, data, and metrics? Second, conditional on such sensitivity, does an approximate method reproduce the distribution of exactly retrained models? If the original model cannot be distinguished from the reference population, approximate/exact proximity is compatible both with genuine removal and with a procedure that leaves the original function essentially unchanged. In that setting the assay cannot support a positive or negative method claim.

The present manuscript reports a bounded retrospective pilot, P001, and the infrastructure used to preserve it. P001 uses one reduced Fashion-MNIST split, one frozen IID deletion manifest, one original seed, and 15 exact-retraining seeds. Its purpose is methodological: to validate exclusion, lineage, immutable evidence, reference-ensemble estimation, and paper generation. The analysis then asks whether P001 was an informative deletion assay. It makes no novelty or successful-unlearning claim.

The distinction between an experimental target and an evaluation convenience is important in this setting. Let $D$ be a training set, $F \subset D$ a requested deletion set, $A$ a stochastic training procedure, and $U$ an approximate update procedure. The usual objects are an original model $M_0 = A(D)$, an updated model $M_U = U(M_0, F)$, and one retrained model $M_R = A(D \setminus F)$. The latter is often used as though it were a fixed oracle. It is not. For a fixed retained dataset, $A(D \setminus F)$ induces a distribution over functions, probabilities, calibration states, and representations. A scientifically cautious comparison therefore asks whether $M_U$ is compatible with that distribution on prespecified probes. It does not ask whether $M_U$ happens to resemble one convenient retraining draw.

This perspective also clarifies why a favorable conventional outcome can be ambiguous. A reduction in forget-set accuracy may reflect meaningful removal, ordinary class confusion, or collateral degradation. Stable retained accuracy may coexist with changed confidence or local behavior. A weak membership attack may be weak because of its threat model rather than because the model is counterfactually faithful. None of these observations is useless; each answers a different question. The manuscript therefore treats their agreement and disagreement as an object of measurement, while declining to collapse them into one unsupported success score.

The project is intentionally staged. Tier-0 and Tier-1 work establishes deterministic data handling, artifact contracts, and deletion invariants. The current P001 release is a bridge between those engineering checks and a future publication-scale study. It is designed to make the transition auditable: what was measured, what was merely designed, what failed, and what remains unmeasured are all separated in the repository and in the text. That separation is less dramatic than a method leaderboard, but it is necessary if later empirical claims are to be interpretable under skeptical review.

The contributions of this release are therefore deliberately narrow:

1. an operational counterfactual evaluation model in which exact retraining is an ensemble rather than one arbitrarily selected network;

2. an audit pipeline that cryptographically binds dataset, split, deletion, training, prediction, metric, figure, and manuscript artifacts;

3. a reproducible P001 case study demonstrating seed-count diagnostics and the need to test deletion identifiability before interpreting method proximity; and

4. a claim boundary and prospective C001 gate that prevent exploratory pilot outcomes from being presented as confirmatory evidence.

The remainder of the paper first positions this scope relative to unlearning, auditing, and evaluation work. It then specifies the counterfactual target, evidence lineage, measurement layers, and statistical boundaries. The results section reports only the completed pilot and its negative assay-identifiability finding. Later sections describe the implications, failure records, limitations, and the conditions that must be met before a publication-grade confirmatory study is allowed to start.

# 2 Related Work

## 2.1 Definitions and counterfactual targets

Certified removal formalizes a distributional relationship between an updated learning system and retraining [4]. Stability-oriented work studies deletion bounds for particular objective classes [7], while generalization-based formulations clarify that correct classification of a deleted IID example need not imply instance-specific memory [8]. These distinctions motivate the present separation between procedural exclusion, empirical behavioral similarity, and a formal certificate. This paper supplies the first two for a pilot and claims neither that the empirical envelope is a certificate nor that matching finite probes establishes erasure.

## 2.2 Efficient unlearning methods

SISA partitions training into shards and slices so that a deletion can trigger partial rather than complete retraining [1]. Gradient-based approaches instead update a trained model, with objectives that may increase loss on forgotten data, recover utility on retained data, or combine teacher and student signals. SCRUB is a representative deep-network method evaluated in the unbounded-unlearning setting [5]. Selective Synaptic Dampening modifies parameters according to estimated importance [2]. The method classes make different structural assumptions and expose different cost profiles; consequently, a single forget-accuracy number is not a sufficient common target.

The P001 implementation includes fine-tuning and naive gradient ascent as simple engineering baselines and a clean-room SCRUB implementation. Their presence validates a common interface but does not establish competitive reproduction. A method enters the confirmatory comparison only after its paper, implementation provenance, license, deviations, and at least one sanity-check behavior are recorded.

## 2.3 Privacy attacks and behavioral evidence

Membership inference tests whether an adversary can distinguish members from non-members using model behavior [10]. It is an important privacy diagnostic, but an attack result is conditional on the adversary, features, reference population, and training data. Chance-level performance for one attack does not demonstrate that the unlearned model behaves like an exact retrain, just as prediction agreement does not rule out privacy leakage. The proposed contradiction matrix therefore reports privacy beside, rather than as a substitute for, predictive, probabilistic, calibration, representation, and local endpoints.

## 2.4 Auditability and benchmark methodology

Auditable definitions motivate retaining the executed algorithm, deletion request, data lineage, and exclusion evidence rather than attempting to infer the entire procedure from outputs [11]. Our artifact design follows this principle by coupling behavioral records with immutable manifests and checkpoint hashes. Deep-unlearning benchmarks compare utility and forgetting objectives [5]; the specific measurement question here is whether those surface criteria can disagree with a population of exact counterfactual models. The contribution sought is therefore an evaluation design and empirical contradiction study, not another isolated method.

Wichert and Sikdar show directly that random forget sets can make an unlearning method difficult to distinguish from retaining the original model, and recommend influential instances for more diagnostic evaluation [12]. MUSE broadens evaluation to six properties and repeated deletion requests for language models [9]. Gao et al. demonstrate that favorable output behavior may coexist with recoverable forgotten information in internal representations [3]. These results substantially narrow the present contribution: neither metric disagreement, representation auditing, sequential evaluation, nor the weakness of random deletion is claimed as new. Lanyon et al. further show that evaluation can depend materially on the original model training seed and that additional unlearning seeds do not generally compensate for a single-training-seed design [6]. Accordingly, varying original training seeds is a required control, not a contribution claimed by this study. The unresolved question is whether a prespecified assay can calibrate approximate-model discrepancies against the natural distribution of independently exact-retrained models and identify contradictions without using approximate-method outcomes to choose the test.

### 2.5 Scope of the present evidence

Fashion-MNIST supplies a modest neural test bed [13]. It is useful for exercising stochastic training and probability-level metrics but cannot establish cross-dataset or architecture-level generality. Digits is reserved for deterministic CI. The staged program moves to stronger vision and tabular settings only after the reference assay and statistical rules are frozen. The complete literature-review and citation-audit records are maintained separately so that unverified recollection cannot silently enter the bibliography.

## 3 Problem Formulation

Let the original training set be $D = D_r \cup D_f$, where $D_f$ is the requested deletion set, $D_r$ is the retained set, and $D_r \cap D_f = \varnothing$. Let $A$ denote a randomized training algorithm and $U$ an approximate unlearning operator. The original, exact, and approximately unlearned models are

$$M_0 = A(D; s_0), \tag{1}$$

$$M_R^{(s)} = A(D_r; s), \tag{2}$$

$$M_U = U(M_0, D_f, D_r; \eta), \tag{3}$$

where $s$ indexes training randomness and $\eta$ contains method-specific randomness and hyperparameters. The exact counterfactual is consequently the population

$$\mathcal{R}(D_f) = \{M_R^{(s)} : s \sim \mathcal{S}\}, \tag{4}$$

not a single model or parameter vector. P001 approximates this population with 15 fixed, independent seeds.

For a probe set $X$ and discrepancy functional $d_j$, exact/exact variation is represented by

$$\mathcal{N}_j = \{d_j(M_R^{(a)}, M_R^{(b)}; X) : a < b\}. \tag{5}$$

This set is an empirical null reference, but its $\binom{k}{2}$ entries are not independent because each trained model appears in multiple pairs. Seed-level or manifest-level resampling is therefore required for inferential procedures.

The original/exact distribution

$$\mathcal{O}_j = \{d_j(M_0, M_R^{(s)}; X) : s \in S\} \tag{6}$$

tests whether deletion produces an observable signal relative to ordinary retraining variation. Only after this reference-sensitivity condition is met do we interpret the approximate/exact distribution

$$\mathcal{U}_j = \{d_j(M_U, M_R^{(s)}; X) : s \in S\}. \tag{7}$$

Counterfactual fidelity is kept multidimensional. A claim can concern predictions, probability distributions, calibration, privacy attacks, intermediate representations, local neighborhoods, robustness under controlled corruptions, retained utility, or computational cost. Agreement on one coordinate does not imply agreement on another.

## 4 Methodology

### 4.1 Audit unit and lineage

The experiment, rather than an individual scalar metric, is the audit unit. Each completed record contains an experiment identifier, configuration hash, dataset identity, explicit split IDs, deletion and retained IDs, architecture, seeds, training and unlearning hyperparameters, runtime environment, metric outputs, and checkpoint and prediction references. Derived artifacts retain hashes of their parents. The intended chain is

source dataset → canonical split → deletion manifest → run configuration → checkpoint → predictions → metrics → statistics → table or figure → manuscript claim.

This chain makes accidental substitution detectable and permits a reader to work backward from a reported number to the inputs that produced it.

The chain is deliberately more granular than a conventional experiment log. Dataset hashes alone do not identify a split; a split hash alone does not identify the deletion request; and a metric file alone does not establish which checkpoint generated it. The audit unit therefore stores both references and content hashes at every boundary where a mutable file could otherwise be substituted. A downstream analysis is required to name its upstream records explicitly. When an output is regenerated from unchanged inputs, the derived file may be reproduced; when an experiment is rerun, its raw record receives a new identifier rather than replacing an old one. This policy supports correction without rewriting history.

## 4.2 Deletion and exclusion invariants

The deletion manifest is generated before exact retraining and identifies stable dataset IDs rather than loader positions alone. The retained manifest must equal the complement of the deletion set within the frozen training split. The implementation asserts pairwise disjointness, complete reconstruction of the training split, and exclusion of every deleted ID from every exact-retraining dataset. A failed assertion invalidates the run; it is not treated as a statistical observation.

These checks distinguish "deleted from the current loader" from "absent from the counterfactual training population." The latter is the relevant property for exact retraining. For every requested deletion, the implementation forms a retained-position array as the set complement of forgotten positions in the already frozen training split. It then verifies that the two arrays are disjoint, that their sorted union reconstructs every training position once, and that their mapped stable IDs have the same properties. Exact-retraining calls receive the forgotten positions as an explicit forbidden set, providing a second boundary against accidental reintroduction. The redundant checks are intentional: they turn a severe scientific error into an observable failure rather than a silent assumption.

## 4.3 Reference construction

All exact replicas use the same retained data, architecture, training schedule, and evaluation probes while varying the declared training seed. The seed list is frozen in the experiment configuration. P001 uses seeds 2001–2015. Pairwise exact/exact comparisons describe natural stochastic variation. For analyses that require uncertainty, the model seed remains the independent unit; the 105 pairwise comparisons are not assigned an independent sample size of 105.

The reference ensemble is not a claim that all stochasticity has been modeled. It conditions on a declared architecture, optimizer, data order policy, preprocessing pipeline, and hardware environment. Changing any of these may change the distribution. Its value is narrower and more practical: it makes the variability induced by the executed training procedure visible, and it prevents a single reference draw from acquiring unjustified authority. When the ensemble is small, the appropriate conclusion is imprecision, not a more confident comparison.

## 4.4 Metric calculations

Prediction disagreement is the fraction of probe examples on which two top-1 predictions differ. For class-probability vectors $p$ and $q$, Jensen–Shannon divergence is

$$\mathrm{JS}(p, q) = \tfrac{1}{2}\,\mathrm{KL}(p\|m) + \tfrac{1}{2}\,\mathrm{KL}(q\|m), \qquad m = \tfrac{1}{2}(p + q), \tag{8}$$

averaged over the probe set. Task and calibration outputs include accuracy, balanced accuracy, macro-F1, negative log-likelihood, Brier score, and expected calibration error with 15 bins. P001 stores additional distances, but only JS divergence and top-1 disagreement were covered by its frozen seed-count diagnostic.

Metrics are computed separately on forgotten, retained, validation, and independent test partitions whenever those partitions are meaningful for a condition. This matters because a global test average can hide a localized disturbance near the deletion set, while a forget-set metric can be dominated by ordinary class-level generalization. Calibration measures are kept separate from accuracy because confidence can change without a top-1 change. Representation comparisons use fixed probes and symmetry-aware summaries; they are not

interpreted as direct neuron-by-neuron correspondence. A metric is therefore a coordinate of fidelity, not a vote in a majority-rule definition of unlearning.

### 4.5 Raw immutability and failures

Completed raw experiment directories are append-only. A rerun receives a distinct identifier and may name a parent experiment. The failure registry retains configuration, environment, exception category, and remediation information, while failed runs are excluded from statistical samples. Derived summaries may be superseded by corrected versions, but their source records remain preserved and corrections are documented.

The failure policy is equally important. An exception, an invalid method mechanism, a missing dependency, an artifact collision, or an unmet frozen acceptance criterion is recorded with its environment and remediation. These events do not become numerical observations, but neither are they deleted as inconvenient development traces. In particular, a method can pass an interface smoke test yet remain ineligible for comparison if its declared mechanism or validation rule fails. The current method-readiness gate makes that status machine-readable, so a blocked method cannot silently return to a launch profile merely because its code remains installed.

## 5 Counterfactual Evaluation Framework

### 5.1 Reference sensitivity before method evaluation

Exact/exact comparisons define an empirical null envelope. Original/exact discrepancy is then compared with that envelope before any approximate-method decision. This ordering is essential: when deleting $D_f$ does not create a detectable shift under metric $j$, a low approximate/exact distance cannot identify whether $U$ removed influence or merely preserved behavior that was already compatible with retraining.

The P001 retrospective analysis summarizes this relation with a separability ratio formed from the original/exact signal and the exact/exact reference scale. Power thresholds were simulated with seed-level summaries. Because this analysis was designed after P001, it is reported as exploratory and cannot retroactively establish a confirmatory threshold.

### 5.2 Fidelity dimensions

The full framework distinguishes the following coordinates:

**Predictive fidelity.** Agreement, disagreement, and class-conditioned errors.

**Probabilistic fidelity.** Distances between complete predictive distributions, including JS divergence and total variation.

**Calibration fidelity.** NLL, Brier score, ECE, and reliability behavior on forgotten, retained, and independent test sets.

**Privacy fidelity.** Difference between attacks against $M_U$ and attacks against the exact population, rather than comparison with chance alone.

**Representation fidelity.** CKA or other symmetry-aware comparisons on a fixed probe set for compatible architectures.

**Local and robustness fidelity.** Behavioral change near deleted observations and under controlled corruptions, without mislabeling corruption tests as adversarial robustness.

**Utility and cost.** Retained/test performance, runtime, device, memory, and cost relative to exact retraining.

### 5.3 Decision states

Composite rules and equivalence margins must be frozen before confirmatory experiments. The current vocabulary distinguishes assay-insensitive (U0), functional failure (U1), functional-only equivalence (U2), strong empirical equivalence (U3), and inconclusive (U4). U3 would remain an empirical, assay-bounded statement rather than certified removal or a legal claim. P001 is assigned U0 retrospectively for its two evaluated output probes; no method receives a confirmatory state.

## 6 Research Questions and Falsifiable Outcomes

### 6.1 Primary question

The publication program asks whether conventional indicators of successful forgetting agree with behavioral consistency relative to models that never encountered the deleted records. Let $C_j(M_u)$ denote the pass/fail decision induced by conventional criterion $j$, and let $E(M_u)$ denote consistency with the exact-retraining reference population. The central estimand is not merely mean method performance, but the frequency and form of the contradiction

$$C_1(M_u) = \cdots = C_q(M_u) = \text{pass}, \qquad E(M_u) = \text{fail}. \tag{9}$$

This definition deliberately permits the criteria to disagree. It does not presume that counterfactual fidelity is one-dimensional or that every discrepancy is practically important.

### 6.2 Research questions

The confirmatory program is organized around five questions. **RQ1** asks whether a deletion intervention is identifiable against exact-training randomness for the chosen regime. **RQ2** asks how often utility, forget-set, privacy, and calibration criteria agree with exact-counterfactual consistency. **RQ3** asks which fidelity dimensions detect contradictions that aggregate accuracy misses. **RQ4** asks whether repeated deletion requests produce cumulative discrepancy. **RQ5** asks how fidelity changes as computational cost is reduced relative to retraining. P001 addresses only the measurement precondition in RQ1; the other questions remain prospective.

### 6.3 Outcome taxonomy

Each evaluated condition has four scientifically distinct outcomes. A *consistent* outcome places the approximate model within the prespecified exact envelope on the primary endpoint. A *contradictory* outcome passes the conventional criteria but fails that envelope. A *concordant failure* fails both. An *uninformative* outcome occurs when reference sensitivity, sample size, run integrity, or precision is inadequate for a decision. The last state is essential: absence of detectable discrepancy is not evidence of equivalence when the assay cannot resolve the intervention.

### 6.4 Falsification criteria

The motivating measurement hypothesis would be weakened if conventional criteria and counterfactual decisions agree across preregistered datasets, deletion regimes, methods, and seeds; if apparent contradictions vanish under independently chosen equivalence margins; or if counterfactual distances add no stable information beyond inexpensive metrics. Conversely, isolated exploratory contradictions are insufficient. A defensible claim requires repeated, integrity-valid observations under a frozen analysis plan and a reference population demonstrably sensitive to the deletion regime.

# 7 Experimental Design

## 7.1 Dataset and split

The completed P001 pilot uses Fashion-MNIST, a ten-class collection of 28-by-28 grayscale fashion images [13]. The standard source contains 60,000 training and 10,000 test images; P001 intentionally uses a reduced deterministic slice to validate the research machinery. The frozen split contains 9,000 training, 1,000 validation, and 2,500 test examples. Dataset acquisition, scaling, split IDs, and the post-scaling dataset hash are recorded. This reduced slice is not presented as publication-scale evidence.

The small slice makes the control path inspectable: a reviewer can reconstruct every split, rerun every exact replica, and inspect each raw record without requiring a large compute allocation. It does not establish that the observed variability, deletion behavior, or method ranking would persist at the scale, augmentation regime, and optimization horizon of a main benchmark. Fashion-MNIST is therefore used here as a bridge between Tier-0 infrastructure checks and a future registered multi-dataset study, not as the empirical basis for a general claim about approximate unlearning.

## 7.2 Deletion request

The deletion policy samples one percent of the frozen training split uniformly without replacement using deletion seed 1009. The resulting immutable manifest contains 90 stable example IDs, leaving 8,910 retained examples. Both lists and their hashes are stored; the retained list is verified as the exact complement of the deletion list. This is an IID deletion, not a class-deletion, high-influence, rare-example, corrupted-label, or memorization stress regime.

IID deletion provides a necessary reference condition because it separates the mechanics of manifest construction and counterfactual retraining from a selection rule that deliberately targets unusual training examples. It is not expected to be the most revealing deletion regime. The experiment schema retains explicit selection interfaces for class-specific, rare-example, clustered, mislabeled, influence-selected, and sequential requests. Those regimes should be introduced as registered factors rather than retrofitted after inspecting the IID pilot, since their selection procedures can change both the difficulty of unlearning and the appropriate interpretation of forget-set metrics.

## 7.3 Model and training

P001 uses the repository's small convolutional architecture with hidden dimension 64. Original and exact models are trained for four epochs with AdamW, batch size 128, learning rate $10^{-3}$, and weight decay $10^{-4}$. Deterministic execution is requested and the device is selected automatically. The original model uses seed 11; the exact population uses seeds 2001–2015. All exact models train on the same 8,910-example retained set.

Holding the architecture, preprocessing, training schedule, and retained set fixed makes initialization and remaining stochastic training effects the intended source of variation within the exact population. Deterministic settings reduce incidental implementation variation within a run, but they do not make a single seed representative of all valid counterfactual trainings. The seed ensemble is therefore a design feature rather than a replication afterthought: its pairwise distances define the reference scale against which an approximate output is assessed.

## 7.4 Approximate engineering baselines

The pilot executes fine-tuning, naive gradient ascent, and SCRUB. Fine-tuning and naive gradient ascent are explicitly simple baselines; they are not described as competitive leading methods. The archived configuration records one unlearning epoch, learning rate $2 \times 10^{-4}$, retain batch size 128, forget batch size 64, retain weight 1.0, forget weight 0.1, and the method-specific controls used by the runner. These choices validate a common interface and are not the product of a publication-scale tuning study.

The distinction between an engineering baseline and a validated comparison method is important. Fine-tuning and naive ascent answer whether the evaluation and artifact pipeline can process simple parameter updates; they do not represent the contemporary method frontier. Literature-backed candidates enter a later

comparative study only after their implementation provenance, licensing, mechanism checks, and tuning protocol have been documented. A method is not elevated to a publication baseline merely because it conforms to the same software API.

### 7.5 Environment and scope

The recorded run uses Python 3.12.3 and PyTorch 2.9.1 with ROCm 7.2 under WSL2 on an AMD Radeon RX 7800 XT. Runtime is measured around the corresponding training operation. R001 includes P001 and separately labeled calibration evidence; none is treated as confirmatory. The C001 design remains an unlocked local draft and may not begin until its decision rules, manifests, and public prospective registration are finalized. Because the first equivalence-margin calibration failed, C001 currently has no admissible equivalence claim.

The accelerator configuration is recorded because time and memory are properties of a specific software–hardware stack, not universal method constants. PyTorch's ROCm backend uses the "cuda" device namespace for compatibility even when CUDA itself is unavailable; the environment record separately reports the ROCm runtime so this naming convention is not misread as an NVIDIA execution claim. CPU execution remains supported for validation and small tests, while future profiles may run unchanged on compatible CUDA, ROCm, Apple Silicon, or cloud accelerators through the device abstraction.

## 8 Statistical Analysis Protocol

### 8.1 Units of analysis and dependence

The model training run is the experimental unit. Predictions on individual examples are repeated observations produced by that unit and are not treated as independent model replicates. Likewise, the $\binom{k}{2}$ exact/exact pairs share models and therefore do not constitute $\binom{k}{2}$ independent samples. Primary uncertainty calculations resample or aggregate at the model-seed level. Example-level resampling may be used for a conditional prediction analysis, but it must not replace uncertainty over training runs.

### 8.2 Reference summaries

For scalar distance $d$, the empirical exact/exact distribution is summarized by its median, dispersion, and prespecified upper quantile. Approximate/exact distances are computed against every valid exact replica. The principal descriptive statistic is the difference between the approximate/exact median and the exact/exact median; a standardized effect and empirical percentile provide scale-relative views. When sample size permits, bootstrap intervals resample exact models rather than treating dependent pairs as IID.

### 8.3 Prospective exact-seed precision rule

The number of exact replicas is selected by a deterministic precision analysis rather than by inspecting approximate-method rankings. Candidate counts are $k \in \{5, 10, 15\}$. For each primary metric, 5,000 repetitions sample whole exact-model seeds without replacement, recompute the induced pairwise-distance mean, and form a 95% interval. A candidate count passes when the interval half-width divided by its median is at most 0.20. The selected count is the first candidate passing for every primary metric. The simulation seed is 73194. Ten exact seeds met the pilot rule for JS divergence in both calibrated full-class conditions. This is a condition-specific pilot result, not a confirmatory or cross-dataset seed-count guarantee.

The publication-scale profile uses a separate fail-closed decision. Its exact-retraining count cannot be smaller than 15, even when a pilot precision curve selects 10. Original-model and approximate-candidate replication are selected independently from $\{5, 8, 10, 15, 20\}$ using validation-only, per-seed endpoint summaries. Each count must satisfy the 20% relative interval-half-width rule in every frozen method–condition group. Test metrics, method rankings, winner labels, and confirmatory outputs are structurally forbidden inputs. At present, the exact-retraining floor is the only supported partial recommendation; the other counts remain unset. A hash-bound decision artifact must reproduce from its cited evidence before any numeric count can enter the main profile.

## 8.4 Equivalence and distinguishability

Difference testing and equivalence testing answer different questions. A failure to reject equality does not establish equivalence. For scalar endpoints with a scientifically defensible margin $\Delta$, two one-sided tests or a compatible confidence-interval rule will assess whether the entire interval lies in $[-\Delta, \Delta]$. Margins must be frozen before confirmatory outcomes are observed and reported in natural units. If no defensible margin exists, the analysis remains estimation-based and does not label models equivalent.

Calibration v1 proposed the condition-matched exact/exact 95th percentile as an upper margin and zero as the lower bound. Its prospectively fixed one-percent residual-class positive control exceeded that margin in only 1/15 class-0 replicas and 0/15 class-5 replicas. The required rate was 80%. The v1 margins are therefore inadmissible. This failure remains part of the evidence record and cannot be repaired post hoc by widening the bound.

The unexecuted v2.1 design separates stochastic reference variability from the scientific meaning of a discrepancy. Let $R_{c,s}$ denote exact model seed $s$ in condition $c$, let $\mathcal{R}_{c,-s}$ be its leave-one-out exact ensemble, and let $T_\rho$ swap the largest and second-largest predicted probabilities on a deterministic class-stratified fraction $\rho$ of a fixed validation probe. For distance $d$, define the seed-level controlled excess

$$\delta_{c,s,\rho} = \text{median}_{r \in \mathcal{R}_{c,-s}}\, d(T_\rho(R_{c,s}), r) - \text{median}_{r \in \mathcal{R}_{c,-s}}\, d(R_{c,s}, r). \tag{10}$$

The primary proposal uses $\rho = 0.01$, which changes exactly 10 of 1,000 validation predictions and, on the balanced Fashion-MNIST probe, exactly one prediction per class. Fractions 0.005 and 0.02 are prespecified sensitivity anchors. Fifteen calibration seeds estimate a candidate margin as the one-sided 95% lower bootstrap bound of the mean primary-control excess JS. Ten disjoint holdout seeds must then yield at least 80% primary effects above the margin, and the holdout medians must increase strictly across the three severities. All inputs are validation-only; test predictions and approximate-method outputs are prohibited.

V2.1 is a design proposal, not a measured result or an approved margin. The choice that one percentage point of class-balanced top-1 disagreement is the smallest scientifically meaningful effect requires an explicit human decision. Until that review and a separate freeze amendment exist, no equivalence test, equivalence label, or main-study launch is authorized.

## 8.5 Multiplicity, exclusions, and failures

No confirmatory family is currently active. The earlier proposed family containing JS divergence and top-1 disagreement is suspended because disagreement was assay-insensitive in the calibrated full-class conditions and the proposed equivalence margin failed. The exploratory method pilot reports continuous estimates without multiplicity-adjusted PASS/FAIL decisions. A later confirmatory family must be frozen prospectively; Holm or another named correction will then be applied within that frozen family.

All planned seeds are retained unless a frozen quality rule excludes the run. Infrastructure failures, numerical failures, and integrity violations are recorded separately and are not statistical observations. Exclusions must be determined without reference to whether a run supports the hypothesis. Missing runs are reported with their configuration and remediation status.

## 8.6 Exploratory versus confirmatory analysis

P001 is retrospective and exploratory. Its seed-count convergence and identifiability diagnostics can inform, but cannot validate, thresholds for C001. The two subsequent calibration pilots are also non-confirmatory. The drafted full-class method pilot uses five original seeds as the resampling units, ten exact references per condition, and 5,000 seed-level bootstrap repetitions. Exact/exact pairs are descriptive null values, not independent observations. No equivalence or contradiction decision is permitted. Before C001 begins, its hypotheses, endpoints, seed-generation procedure, correction family, stopping rule, and failed-run policy must be timestamped. Any later change is an amendment, and the affected result is labeled exploratory or post hoc.

### 8.7 Sensitivity analyses

Planned robustness checks vary the exact-envelope quantile, distance aggregation, seed subsets, and calibration binning while preserving the primary analysis. A conclusion that depends on one arbitrary bin count or one exact replica is considered unstable. For local analyses, neighborhood definitions are frozen independently of the unlearning outcome to avoid selecting regions that maximize observed discrepancy.

## 9 Sequential Deletion and Scaling Protocol

### 9.1 Stateful deletion process

Sequential unlearning is modeled as a trajectory rather than a collection of unrelated single deletions. Starting from $M^{(0)}$, request $F_t$ produces approximate state $M_u^{(t)} = U(M_u^{(t-1)}, F_t)$. Its counterfactual reference population is independently trained on

$$D_r^{(t)} = D \setminus \bigcup_{s=1}^{t} F_s. \tag{11}$$

The cumulative deletion manifest is immutable at every step, and all previous forgotten identifiers remain excluded. Exact models are not initialized from the approximate state.

### 9.2 Trajectory endpoints

At each step the study records utility, probability divergence, calibration, privacy, representation, local behavior, and cost. Drift is analyzed both absolutely and relative to the contemporaneous exact envelope. Candidate patterns include monotone accumulation, nonlinear failure, plateau, and recovery, but none is assumed. The area above the exact envelope may be reported as a descriptive trajectory summary only after its weighting and normalization are fixed.

### 9.3 Deletion regimes

The architecture supports random, high-loss, class-specific, rare-example, representation-clustered, mislabeled, influence-selected, and sequential selections. Selection is performed from frozen eligible indices and produces a hashed manifest before unlearning. High-loss and influence-based policies must calculate their scores from an explicit model checkpoint and may not use test labels. Corruption experiments preserve both clean and corrupted labels so that removal of harmful data is distinguishable from ordinary IID deletion.

### 9.4 Compute–fidelity frontier

Cost is measured jointly with fidelity rather than presented as an unconditional speedup. For method $a$, retraining-relative cost is

$$\rho_a = \frac{T_a}{\operatorname{median}_i T(M_{r_i})}, \tag{12}$$

with hardware, device, synchronization policy, peak memory, and measurement boundary recorded. Pareto comparisons identify methods for which no alternative is both cheaper and more faithful. Wall-clock results remain environment-specific and are not extrapolated across accelerators without measurement.

### 9.5 Escalation across datasets and architectures

Digits remains a deterministic infrastructure test. Fashion-MNIST provides a low-cost neural pilot. CIFAR-10 is intended to test stronger convolutional architectures, CIFAR-100 increases class granularity, and a tabular dataset tests whether observations depend on vision-specific representations. Dataset inclusion is tied to a generalization question, not to benchmark count. The same manifests, method interface, raw schema, and analysis pipeline are retained across CPU, CUDA, ROCm, Apple accelerators, and remote hardware; only device adapters and environment records vary.

# 10 Metric Specification and Contradiction Matrix

## 10.1 Task utility

Accuracy is reported on retained, forgotten, validation, and test partitions with the partition identity explicit. Balanced accuracy and macro-F1 are added when class balance or deletion policy makes ordinary accuracy misleading. Utility is always compared with the exact-retraining population: retaining the original model's accuracy is not sufficient if exact retraining induces a materially different decision function.

## 10.2 Predictive and probabilistic fidelity

For probe set $S$, prediction agreement between models $a$ and $b$ is

$$A_S(a, b) = |S|^{-1} \sum_{x \in S} \mathbf{1}\{\arg\max_c p_a(c \mid x) = \arg\max_c p_b(c \mid x)\}. \tag{13}$$

Agreement discards confidence information, so it is paired with per-example Jensen–Shannon divergence. Class-conditional summaries detect discrepancies hidden by majority classes. The study retains the distribution of example-level values and the model-level aggregate; only the latter supplies independent training-run replication.

Distance direction also matters. A low divergence from the original model is not generally a fidelity objective, because the original model did train on the forgotten data. The relevant comparison is approximate-to-exact relative to exact-to-exact variation. Reporting both agreement and divergence avoids a false sense of consistency when labels agree but confidence changes substantially, while retaining class-conditioned summaries makes a single aggregate less likely to be dominated by easy classes.

## 10.3 Calibration fidelity

Negative log likelihood evaluates probability assigned to the observed class, and the Brier score measures squared probability error over all classes. Expected calibration error is included with binning fixed in the protocol, accompanied by reliability curves so that one scalar does not conceal the direction of miscalibration. These quantities are computed separately on forgotten, retained, and independent test data. Calibration fidelity means proximity to the exact population, not merely a lower calibration error than the original model.

## 10.4 Forgotten and local behavior

Forget-set accuracy is descriptive for IID deletion and is not interpreted as direct proof of removal. Local analysis evaluates changes among neighbors of forgotten examples in input or frozen representation space. Distance bands, same-class restrictions, probe checkpoint, and tie handling are specified before outcome inspection. The resulting curve asks whether the intervention creates collateral behavioral disturbance that decays with distance from the deletion request.

Local probes require particular care because the chosen neighborhood can itself determine the apparent result. Input-space distance may be poorly aligned with visual similarity, whereas representation-space distance depends on a checkpoint whose training data and layer must be declared. The protocol therefore treats neighborhood construction as part of the experimental condition. Exploratory visualizations can generate hypotheses, but they cannot be promoted to confirmatory evidence after the radius, embedding, or examples have been selected by inspecting the observed effect.

## 10.5 Privacy fidelity

Membership inference is evaluated relative to exact retraining under the same attacker and data construction. Reported quantities include attack AUROC, advantage, and calibration where applicable. An approximate model is not credited merely because one attack is near chance; its privacy behavior must be considered against both exact replicas and alternative attacks. P001 does not contain a validated privacy result, so no privacy conclusion is drawn from it.

### 10.6 Representation fidelity

Compatible architectures are compared on a fixed probe set using centered kernel alignment or another symmetry-aware method. Layer correspondence is declared in advance. Direct neuron-by-neuron comparison is avoided because hidden-unit permutations can preserve the function. Linear probes may characterize class separability but do not by themselves establish representational equivalence.

### 10.7 Robustness fidelity

Controlled corruptions test whether approximate and exact models respond similarly away from the clean empirical distribution. Candidate transformations include Gaussian noise, blur, brightness change, and occlusion with frozen severity levels. These are corruption tests, not adversarial-robustness claims. Clean and corrupted outcomes are linked to the same model run so that uncertainty is propagated at the seed level.

### 10.8 Contradiction matrix

For every condition, the processed record contains separate decisions for forget behavior, retained utility, privacy, calibration, prediction distribution, representation, and exact counterfactual consistency. Each cell is `pass`, `fail`, `inconclusive`, or `not assessed`; missing evidence is never converted to a pass. A contradiction is counted only when the conventional criteria and counterfactual criterion were frozen, evaluable, and integrity-valid. The matrix preserves component values so that later aggregation cannot hide conflicting dimensions.

The matrix is not a composite score. Its purpose is diagnostic: it reveals whether a method that satisfies one operational objective is nevertheless outside the counterfactual reference envelope on another observable. Any future aggregate would need a defensible utility model, sensitivity analysis, and a demonstration that it does not conceal these conflicts. Until then, an `inconclusive` cell is scientifically preferable to a forced binary success label.

The next full-class method pilot does not meet those prerequisites. It therefore emits a continuous exploratory comparison rather than a contradiction count: every original-seed method result retains its conventional metrics, median distance to the exact ensemble, and location within the dependent exact/exact null. No cell is converted to pass or fail, and no method is labeled equivalent.

## 11 Scientific Artifact Architecture

### 11.1 Experiment manifests

Every planned run is generated from a machine-readable condition containing dataset and version, architecture, training seed, deletion strategy and fraction, method and hyperparameters, exact seeds, metrics, and expected hardware class. Reduced, main, and extended profiles make the size of the experiment explicit before execution. Expansion calculates the number of trainings, approximate device-hours, checkpoint volume, and result count, preventing an accidental factorial explosion.

### 11.2 Canonical data lineage

The lineage chain is

$$\text{source dataset} \to \text{preprocessing} \to \text{canonical split} \to \text{deletion manifest} \to \text{run configuration} \to \text{checkpoint and predictions} \to \text{metrics} \to \text{statistics} \to \text{figure or table} \to \text{claim}.$$

Each edge carries a stable identifier or cryptographic hash. Split indices and deletion indices are archived rather than reconstructed from prose. Dataset validation records schema, counts, class balance, duplicate rate, missing values, label validity, overlap, and source integrity in both human- and machine-readable forms.

Lineage is deliberately more granular than an experiment name. A result that only records "Fashion-MNIST, seed 11" cannot establish which preprocessing revision, split, or deletion request created it. The stored chain makes counterfactual exclusion auditable: the exact-training manifest can be compared directly

with the forgotten-ID manifest rather than inferred from control flow. This also permits a later reviewer to distinguish a model trained under a new preprocessing version from a genuine rerun of an earlier condition.

### 11.3 Append-only raw records

A completed run receives a unique identifier, timestamp, configuration hash, Git commit, environment snapshot, device, runtime, seed, metrics, and checkpoint references. A rerun creates a new record and may point to its parent; it never replaces the earlier payload. Raw predictions remain available for metric recomputation. Processed summaries are rebuildable and are kept physically distinct from raw evidence.

Immutability is not a claim that files are technologically impossible to alter; it is an operational contract. New execution creates new evidence, and validation records the configuration and content identifiers required to detect substitution or accidental overwrite. Derived artifacts may be regenerated when analysis code changes, but the derivation must identify the raw inputs and software revision used. This separation allows an analysis bug to be corrected without rewriting the history of the run that exposed it.

### 11.4 Acceptance gates

A run enters the paper dataset only when dataset and split hashes match, retained and forgotten partitions are disjoint, every forgotten identifier is absent from exact training, training completed, predictions exist, metrics validate, metadata is complete, and unexplained non-finite values are absent. Gate success means that these properties were checked; it does not mean that the unlearning method succeeded. Failed runs remain in a separate registry with category, exception, configuration, batch impact, and remediation.

The acceptance gate is intentionally conservative. It cannot prove that a method achieved unlearning, but it can reject records whose numerical values would otherwise invite an unsupported interpretation. For example, a completed training job is insufficient if its forgotten examples leaked into the exact-retraining dataset, if predictions for a required partition are absent, or if a metric silently contains non-finite values. Failed runs are not pooled as observations, yet their presence matters for operational reproducibility and for assessing whether a stated experimental matrix was actually completed.

### 11.5 Claim–evidence graph

Substantive quantitative claims are represented as nodes that point to experiment IDs, processed results, statistical analyses, and figure or table outputs. Release validation rejects a verified claim if any referenced artifact is absent or hash-inconsistent. This mechanism reduces manual transcription and distinguishes measured statements from interpretation, hypotheses, and limitations. Headline LaTeX numbers and tables are generated from validated artifacts rather than copied into prose.

### 11.6 Release and independent reproduction

The release bundle contains source, locked configuration, environment records, immutable evidence, analysis programs, generated values, figures, tables, manuscript source, and checksums. Level-A reproduction starts from archived predictions; Level-B reproduction repeats acquisition and training; Level-C replication independently implements the written protocol. These levels are reported separately because a reproducible script does not by itself demonstrate independent scientific replication.

### 11.7 Private development disclosure record

Substantial automated assistance is recorded privately by activity class—implementation, orchestration, analysis, visualization, and drafting—so that the human author can comply with a venue's disclosure policy. Prompt transcripts and irrelevant scratch traces are not scientific provenance and need not be public. Dataset lineage, executed configurations, failures, numerical evidence, and manuscript claim links are scientific provenance and are retained regardless of the development tools used.

# 12 Pilot Results

## 12.1 Integrity outcome

P001 completed all ten recorded integrity checks. Table 1 is generated from the immutable raw record; the overall gate passed while the record remains marked pilot and not publication-eligible.

This outcome is procedural rather than substantive. It establishes that the recorded run contains the required manifests, integrity checks, metadata, and metric outputs in a form that can be traced through the analysis pipeline. It does not validate the sensitivity of a particular deletion assay, the competitive quality of an approximate method, or a counterfactual-equivalence claim. Keeping these interpretations separate is essential: an auditable run can still be scientifically uninformative for the question it was intended to test.

Table 1: P001 integrity-gate outcomes generated from the raw experiment record.

| Gate | P001 status |
|---|---|
| All Numeric Outputs Finite | PASS |
| Checkpoints Present And Hashed | PASS |
| Configuration Hash Present | PASS |
| Exact Exclusion Assertion | PASS |
| Forget Retain Disjoint | PASS |
| Multiple Exact Seeds | PASS |
| Partition Reconstructs Train | PASS |
| Process Completed | PASS |
| Raw Predictions And Labels Present | PASS |
| Recognized Maturity | PASS |
| Runtime Complete | PASS |

## 12.2 Exact-retraining population

The 15 exact replicas exhibit nonzero variability on all reported splits. Table 2 reports per-seed ranges and means. Forget-set accuracy ranges from 0.8222 to 0.9222, while test accuracy ranges from 0.8096 to 0.8504. These ranges illustrate why neither one exact model nor one absolute forget-set value is an adequate target.

The displayed ranges are descriptive summaries of this finite ensemble, not confidence intervals for a population of all possible trainings. Their main role is to make the comparison target explicit. An approximate model should be considered in relation to the dispersion among models trained on the retained dataset, rather than treated as successful or unsuccessful simply because it differs from one selected retraining seed. Later studies will need to report uncertainty for the envelope itself and assess whether its estimates are stable under additional seeds and independent original models.

## 12.3 Frozen seed-count diagnostic

The frozen $k = 10 \to 15$ diagnostics changed by 4.48% for mean Jensen–Shannon divergence and 3.14% for prediction disagreement. Both values fall below the predefined 10% pilot threshold. The ensemble therefore passed the stated pilot convergence diagnostic for these two metrics. This does not establish that 15 seeds are sufficient for other datasets, architectures, deletion regimes, or metrics, and the threshold is not an equivalence margin.

The diagnostic was intentionally limited to two inexpensive predictive distances. It asks whether enlarging the finite reference ensemble changed those descriptive means by less than a predeclared pilot tolerance; it does not demonstrate convergence of all fidelity dimensions. In particular, calibration, local behavior, membership-inference outcomes, and representation measures may require different seed counts. The procedure also did not adaptively stop once favorable values were observed, which preserves its role as a fixed engineering check rather than an outcome-driven sample-size decision.

Table 2: Metric ranges across 15 independently exact-retrained models. NLL is negative log-likelihood and ECE uses 15 bins.

| Split and metric | Minimum | Mean | Maximum |
|---|---|---|---|
| Forget Accuracy | 0.8222 | 0.8741 | 0.9222 |
| Forget NLL | 0.3426 | 0.4040 | 0.4750 |
| Forget Brier | 0.1568 | 0.1959 | 0.2356 |
| Forget ECE | 0.0510 | 0.0839 | 0.1240 |
| Retain Accuracy | 0.8263 | 0.8509 | 0.8695 |
| Retain NLL | 0.3579 | 0.4078 | 0.4587 |
| Retain Brier | 0.1861 | 0.2128 | 0.2410 |
| Retain ECE | 0.0086 | 0.0248 | 0.0447 |
| Test Accuracy | 0.8096 | 0.8373 | 0.8504 |
| Test NLL | 0.4145 | 0.4584 | 0.5111 |
| Test Brier | 0.2116 | 0.2333 | 0.2628 |
| Test ECE | 0.0120 | 0.0204 | 0.0422 |

## 12.4 Approximate-method descriptives

Table 3 records the executed method runtimes and accuracies. The values confirm that the common method and metric interfaces executed, but they are not a ranking. In particular, high accuracy on the IID forget set does not demonstrate that the deleted records remain memorized: exact models that never trained on those records also achieve substantial forget-set accuracy.

This distinction is especially consequential for semantically ordinary IID examples. A counterfactual model can classify a deleted garment correctly because it learned related structure from retained examples, while an unlearned model can reduce accuracy through collateral damage rather than faithful removal of the deleted example's influence. The descriptive table should therefore be read as evidence that the baseline interfaces ran and produced comparable artifacts, not as evidence that one baseline is more effective than another.

Table 3: Descriptive P001 outputs for approximate methods. These values are not interpreted as confirmatory unlearning outcomes because the deletion assay was retrospectively U0.

| Method | Seconds | Forget acc. | Retain acc. | Test acc. |
|---|---|---|---|---|
| Fine-tuning | 0.27 | 0.900 | 0.865 | 0.845 |
| Naive gradient ascent | 1.15 | 0.900 | 0.863 | 0.849 |
| SCRUB | 1.87 | 0.922 | 0.869 | 0.850 |

## 12.5 Retrospective identifiability analysis

A retrospective separability analysis compared the original model with the exact ensemble. The separability ratios were 0.442 for JS divergence and 0.577 for prediction disagreement. Both were below their exploratory power-derived thresholds, yielding U0 (assay-insensitive). Consequently, the pilot does not determine whether any approximate method forgot the IID-deleted examples. No confirmatory method result is reported.

U0 is an assay statement, not a statement that deleted examples had no effect on training. It means that, under the archived perturbation construction and selected distances, the original-versus-exact contrast was not sufficiently separated to support the intended decision. A stronger deletion regime, a different probe distribution, a longer training schedule, or a different architecture may yield a measurable contrast; those are hypotheses for prospectively specified studies rather than explanations that can rescue the current pilot.

### 12.6 Prospective class-conditioned assay calibration

A separate prospective pilot used five original seeds and 15 exact-retraining seeds per condition, without evaluating approximate methods. Full deletion of Fashion-MNIST class 0 and class 5 produced JS-divergence separability ratios of 1.958 and 2.275, exceeding their simulated minimum detectable ratios of 1.45 and 1.60. Half-class deletion and every tested disagreement, CKA-distance, and MMD endpoint remained assay-insensitive. A model-level precision simulation selected ten exact seeds for JS under the frozen 20% interval rule in both full-class conditions.

### 12.7 Prospective equivalence-margin calibration

A second frozen pilot tested whether the condition-specific exact/exact 95th percentile could detect a controlled violation in which 1% of the forbidden class remained in training. The candidate margins were 0.031825 for class 0 and 0.034819 for class 5. Detection probabilities were only 6.7% and 0.0%, respectively, below the prospectively required 80% in both cases. The candidate margins are therefore inadmissible. Larger contamination levels were designated diagnostics and cannot rescue the failed primary rule. No approximate-method equivalence claim is made.

### 12.8 Runtime

Table 4 separates original training, aggregate exact retraining, and the aggregate approximate-method execution. Runtime values characterize this software and hardware configuration and are not universal performance measurements. Speed alone has no scientific meaning without an interpretable fidelity comparison.

The recorded timing is deliberately narrow: it measures the wall-clock scope defined by the runner on one ROCm environment. It is not an estimate of energy consumption, peak memory, cloud cost, or accelerator-independent throughput. A later compute–fidelity frontier must standardize timing boundaries, preserve device and driver metadata, and report both the cost of the approximate update and the cost of constructing the exact-retraining reference population needed to evaluate it.

Table 4: Recorded P001 runtime on the RX 7800 XT/ROCm environment.

| Operation | Seconds |
|---|---|
| Original training | 9.59 |
| 15 exact retrainings, aggregate | 17.46 |
| Approximate methods, aggregate | 3.29 |

## 13 Analysis

### 13.1 Dependence among exact/exact pairs

The pilot convergence analysis bootstraps independently trained exact replicas rather than assigning an independent sample size of $\binom{15}{2}$. Phase 3 power simulation likewise uses one summary per exact-model seed. P001 contains only one original-model seed, so its retrospective U0 classification motivates but cannot freeze the prospective assay threshold. All numerical values in the preceding section are generated from hash-bound JSON artifacts by the paper build.

### 13.2 Why the convergence and sensitivity questions differ

The convergence diagnostic asks whether adding five exact replicas materially changes two estimated reference summaries. The separability analysis asks whether deletion produces a signal that exceeds normal exact-retraining variability. A stable estimate can describe a small or uninformative signal precisely; convergence therefore does not imply assay sensitivity. P001 passed the former diagnostic and failed the latter retrospective test.

### 13.3 Interpretation of forget-set accuracy

For an IID classification deletion, the target label and class-level visual structure are shared across many retained examples. An exact model trained without a particular image can still classify that image correctly. Accordingly, the relevant comparison is between the distribution of $\mathrm{Acc}(M_U, D_f)$ and the distribution of $\mathrm{Acc}(M_R, D_f)$, not an expectation that forget accuracy approach zero. P001's exact replicas span 0.8222–0.9222 forget accuracy; the approximate values in Table 3 lie within that descriptive range. Because the output assay is U0, this observation supports neither successful nor failed removal.

### 13.4 Claim discipline

The result graph distinguishes measured facts from interpretations. Integrity-gate status, seed counts, metric values, and runtimes are experimental outputs. The U0 label is a retrospective analysis with an explicit one-original-seed limitation. The need for stronger regimes is a design inference. Certified deletion, privacy equivalence, generalization, and method superiority remain unsupported. This classification is encoded in the claim graph used by the paper gate.

## 14 Discussion

### 14.1 What P001 establishes

P001 establishes that the repository can freeze a deletion request, construct an exact retained complement, prevent deleted observations from entering exact retraining, execute multiple independent replicas, store raw predictions and checkpoints with hashes, derive reference statistics, and generate manuscript values from those artifacts. It also shows that stochastic variation is empirically visible even in a small deterministic pilot. These are infrastructure and measurement findings, not evidence that an approximate method satisfies a general definition of unlearning.

### 14.2 Identifiability before equivalence

The pilot exposes an identifiability problem rather than a method ranking. Ordinary IID deletions may leave the trained function almost unchanged relative to ordinary retraining variation. In such a regime, approximate/exact proximity is compatible both with genuine removal and with doing essentially nothing. Stronger class-conditioned, influential, and controlled-memorization regimes are therefore required. This interpretation remains restricted to assay design and does not establish the behavior of any unlearning method.

This ordering matters for equivalence testing. An equivalence interval contained within a margin can support similarity only for the measured endpoint and only if the endpoint is capable of detecting the relevant intervention. Otherwise scientifically different procedures may satisfy the same surface criterion. C001 therefore requires both a reference-sensitivity gate and prospectively frozen equivalence margins.

The class-conditioned calibration makes this distinction concrete. JS divergence detected the original-versus-full-deletion intervention for two classes, yet the exact/exact 95th percentile did not detect the prospectively frozen one-percent residual-class violation. Identifiability of a large intervention is therefore not sufficient to validate a tight equivalence margin. The failed margin is retained as a negative design result rather than widened after observing the calibration.

### 14.3 Why multiple exact replicas matter

Across the 15 exact seeds, test accuracy varies by more than four percentage points and forget accuracy by ten percentage points. Selecting one replica can therefore change the apparent distance or utility cost assigned to an approximate method. An ensemble exposes this dependence and supports empirical percentiles, bootstrap intervals, or equivalence tests at the model-seed level. It does not make the finite ensemble identical to the full training distribution, but it makes uncertainty visible.

### 14.4 Auditability as a methodological property

Output similarity alone cannot reveal which data were used or whether the declared algorithm was actually executed. The repository therefore couples behavioral evidence with procedural evidence: immutable manifests, exclusion assertions, configuration and artifact hashes, append-only raw records, failure metadata, and claim-to-artifact links. This does not solve every auditing problem, but it narrows the gap between a reported result and the executable process that produced it.

### 14.5 Efficiency conditional on fidelity

The aggregate approximate-method runtime is smaller than aggregate exact retraining in P001, but the comparison is a small hardware-specific observation. Computational savings become scientifically meaningful only after an assay can distinguish the relevant counterfactual and a method's fidelity is characterized. The planned compute–fidelity frontier therefore reports cost jointly with multiple fidelity dimensions rather than as a standalone speedup claim.

### 14.6 Implications for C001

The next study should retain class-conditioned deletion as a distinguishability stress regime while treating equivalence as unresolved. Influence-selected observations, controlled memorization, and corrupted-label removal remain candidate regimes. Multiple original seeds are needed to estimate original/exact variation rather than conditioning on one original model. Metric selection, uncertainty reporting, multiplicity control, exclusions, and stopping rules must be registered before confirmatory training begins.

## 15 Limitations and Threats to Validity

### 15.1 External validity

The current evidence covers one reduced Fashion-MNIST split, one small CNN, one IID deletion manifest, and one original-model seed in P001. It does not establish behavior on the full Fashion-MNIST training set, CIFAR, tabular data, larger architectures, sequential requests, structured deletions, or different optimization regimes. The multi-original and controlled-contamination assays are separate pilots rather than confirmatory evidence. Their full-class results cover only classes 0 and 5.

### 15.2 Construct validity

The frozen convergence rule covers JS divergence and prediction disagreement only. It does not validate calibration, privacy, representation, local, relearning, or corruption probes. Representation and parameter similarity do not uniquely determine functional influence. Membership attacks are incomplete adversaries whose outcomes depend on threat model, features, and attack training data. No aggregate score has been justified.

### 15.3 Statistical validity

P001 contains 15 exact seeds but only one original seed and one deletion manifest. The retrospective separability analysis was not preregistered and cannot supply confirmatory error control. Pairwise exact distances are dependent. The frozen 10% convergence threshold is an engineering diagnostic, not a power guarantee or equivalence margin. The class-conditioned precision simulation selected ten exact seeds for JS, but only under the observed finite ensemble and frozen pilot rule. The controlled one-percent contamination calibration rejected the proposed exact/exact-percentile equivalence margins. No claim is made that 10 or 15 seeds, or any numerical margin, transfers beyond these pilot conditions.

## 15.4 Implementation validity

Fine-tuning and naive gradient ascent are simple engineering baselines. Although SCRUB is literature-backed, P001 is not a full reproduction or competitive hyperparameter study. Post-pilot objective auditing found an architecture-specific validity failure: the deterministic small CNN gives the identically initialized student and teacher identical logits at SCRUB's first max-step. The KL term therefore supplies no model-derived ascent signal at that point; numerical residuals amplified by Adam are not accepted as evidence of the published forgetting mechanism. The archived SCRUB output is retained as an immutable interface smoke test but is invalid as method reproduction or competitive evidence. New runs fail closed unless the initial max-step signal is validated. The environment records a single AMD/ROCm/WSL2 execution; bitwise cross-hardware reproducibility and independent laboratory replication have not been demonstrated.

## 15.5 Privacy and legal scope

Approximate unlearning does not by itself establish legal compliance or demonstrate the absence of private information. Empirical proximity to a finite exact-retraining ensemble is not a certificate of erasure. Fashion-MNIST is non-sensitive benchmark data; conclusions about operational deletion requests involving personal information would require a separate threat model, governance process, and privacy analysis.

## 15.6 Measurement and selection threats

The meaning of a fidelity result depends on the probe distribution and the deletion rule. Aggregate test metrics can be insensitive to a localized change near forgotten examples, while a forget-set metric can be confounded by ordinary class generalization. Likewise, high-loss, class-conditioned, clustered, or corrupted-label selections may create a substantially different counterfactual target than IID selection. The current work does not claim that a single collection of metrics is sufficient. It supplies a framework for reporting disagreement among metric families and requires future studies to preserve the selection algorithm, its seed, and its provenance alongside the deletion IDs.

Finite probe sets also limit what can be observed. Prediction agreement and JS divergence are computed on enumerated partitions; two models can agree on those points and differ elsewhere. Conversely, finite exact ensembles can make an approximate output appear unusual because the reference distribution is itself estimated imprecisely. Bootstrap, permutation, and equivalence procedures in a confirmatory protocol must account for this estimation problem, dependent pairwise comparisons, and the multiplicity introduced by methods, datasets, deletion regimes, and metric families.

## 15.7 Temporal scope and method readiness

This manuscript reports the completed P001 pilot and its associated calibration material. It is not a retrospective container for every later development run in the repository. Mechanism-validation failures and tuning failures are retained in the failure registry so that they cannot disappear from the scientific record, but a failed development method is not converted into a statistical observation or a headline result. Before a literature-backed method enters a comparative claim, its provenance, license, implementation deviations, mechanism checks, and prospectively declared tuning policy must be reviewed.

The same restriction applies to the planned main study. A multi-original-seed assay, stronger deletion stressors, sequential trajectories, and additional datasets are necessary to distinguish a pilot measurement issue from a reproducible phenomenon. Until those conditions are frozen and executed, language about evaluation contradictions remains a research question rather than an empirical conclusion.

## 15.8 Publication status

R001 is a locally verified release candidate. At the time of writing it has no public OSF, Zenodo, GitHub release, or preprint identifier, and the author metadata still requires human verification. These omissions are recorded rather than filled with invented identifiers.

Publication readiness is therefore a release gate, not a document-length judgment. It requires verified citations, a claims–evidence graph for every substantive quantitative statement, complete primary experiments

under a frozen plan, documented failures, and a clean-environment reproduction of at least one flagship result. The absence of a public identifier is not a temporary formatting defect: no DOI, registration number, or archival URL should be inserted until the corresponding external action has actually occurred.

# 16 Reproducibility Statement

Research Release R001 provides source code, dataset acquisition and validation procedures, dataset and split hashes, the immutable deletion and retained manifests, all 15 exact seeds, model and method configurations, environment metadata, checkpoint and prediction references, raw and processed metrics, quality-gate output, F002 source data and generation code, and artifact checksums. Level A reproduction (predictions to metrics, tables, figures, and paper values) is supported. Level B reproduction (dataset acquisition through training and analysis) is the target of the R001 package and remains hardware- and software-sensitive. Level C independent reimplementation has not been demonstrated.

**Level A: analysis reproduction.** Starting from archived predictions, a researcher can recompute metrics and seed summaries, regenerate validated tables and F002, regenerate LaTeX commands, and rebuild the manuscript. The release checksum manifest verifies every included payload before analysis.

**Level B: experiment reproduction.** Starting from Fashion-MNIST, the configuration specifies acquisition, split seed, limits, deletion manifest, model, training schedule, original seed, exact seeds, approximate methods, and evaluation parameters. A reproduction creates a new immutable experiment record; it must not overwrite P001. Runtime and floating-point results may vary across drivers and accelerators, so acceptance depends on declared tolerances and invariant checks rather than exact wall-clock or bitwise identity.

**Level C: independent replication.** An independent group would implement the written protocol without relying on hidden local state and assess whether the methodological conclusion persists. No such replication has been completed. The manuscript consequently distinguishes reproducibility infrastructure from independently reproduced evidence.

# 17 Data and Artifact Availability

The local R001 release bundle is machine-verifiable. No public repository release, OSF DOI, Zenodo DOI, or preprint identifier existed at build time; those fields remain explicitly unassigned. Fashion-MNIST is obtained from its cited source. The archived deletion manifest is identified by its complete SHA-256 value in the machine-readable reproduction manifest and release checksum inventory. Identifiers will be inserted only after the corresponding public services assign them.

# 18 Research Transparency

This manuscript distinguishes pilot, exploratory, and confirmatory analyses. P001 was completed before public registration and is therefore described as retrospective. C001 is specified as a prospective confirmatory study and must be registered before its first training run. Automated systems assisted with code, organization, analysis tooling, and drafting; empirical measurements and hashes are generated by the repository, while the human author remains responsible for scientific decisions, interpretation, originality, citation accuracy, and submission-policy compliance.

# 19 Broader and Ethical Considerations

Empirical similarity to a finite retraining population is not proof that personal data has been erased, nor is it a legal compliance certificate. Attack results depend on the adversary and data composition. Releases must avoid exposing deleted or sensitive records.

# 20 Conclusion

P001 demonstrates a reproducible pipeline and motivates reference separability as a precondition for empirical unlearning claims. Its one-percent IID deletion is assay-insensitive under the evaluated output metrics. A prospective class-conditioned pilot identified full-class deletion with JS divergence, while the subsequent controlled- contamination pilot rejected the proposed exact/exact-percentile equivalence margin. The current evidence therefore supports a bounded distinguishability study, not functional, privacy, representation, or utility equivalence. C001 remains blocked pending a revised, prospectively locked decision protocol.

# A Reproducibility Appendix

P001 is identified by 'fashion-cnn-random-01pct-schema2-rocm-pilot-v1'; its exact seeds are 2001–2015 and its deletion manifest is archived under 'experiments/P001'. Original training took 9.59 seconds and all exact retrainings took 17.46 seconds on the recorded RX 7800 XT/ROCm environment. These device measurements are not general performance estimates. The repository stores complete configurations, failures, environment metadata, prediction/checkpoint hashes, and figure source provenance.

## A.1 Frozen identifiers

The original training seed is 11, split seed is 1729, and deletion seed is 1009. Exact training uses seeds 2001 through 2015. The deletion manifest contains 90 IDs and the retained complement 8,910 IDs. Their full SHA-256 values are stored in `reproduction.yaml` and the R001 checksum inventory rather than manually duplicated throughout the prose.

## A.2 Optimization details

Original and exact training use AdamW for four epochs with batch size 128, learning rate $10^{-3}$, and weight decay $10^{-4}$. The unlearning interface uses Adam, learning rate $2 \times 10^{-4}$, one epoch, retain batch size 128, and forget batch size 64. The raw configuration remains authoritative for method-specific alpha, gamma, loss weights, temperature, step limits, and any future correction.

## A.3 Quality-gate semantics

A PASS means the corresponding property was checked for the archived execution. It does not mean the model successfully unlearned. Publication eligibility is separately encoded and is false for P001. Failed experiments are metadata but are not included as statistical observations.

## A.4 Paper generation

The build reads the immutable raw result and processed convergence and separability records, writes `results/publication/paper-results.json`, generates tables and LaTeX commands, validates citations and claim links, builds the manuscript and supplement, and records hashes in `paper/provenance.json`. The R001 release builder copies these outputs, creates a source archive, computes SHA-256 for each payload, and refuses to overwrite an existing release directory.

## A.5 Prospective boundary

C001 is not authorized to run. Two local calibration pilots were completed before C001: the 50-model class-conditioned assay and the 95-model controlled-contamination margin test. Both remain pilot evidence. Before the first confirmatory training job, a public prospective registration must freeze its hypotheses, experimental units, datasets, deletion regimes, manifest generation, training and exact seed counts, primary and secondary metrics, distinguishability rules, correction family, exclusions, failure policy, and stopping rule. Any equivalence claim requires a new admissible margin. Changes after registration must be timestamped and labeled exploratory or confirmatory-impacting.

### A.6 Prospective margin-calibration artifacts

The failed v1 calibration, the unexecuted global-selection v2 draft, and the class-stratified v2.1 alternative are retained as distinct versioned objects. The v2.1 package contains a machine-readable protocol, a checksummed input plan, exact calibration and holdout seed lists, hashes for 30 reusable checkpoints, planned identifiers for 20 new exact trainings, a guarded validation-prediction exporter, and a checkpoint preflight report. The preflight verifies checkpoint hashes, finite tensor state, strict SmallCNN schema compatibility, and the frozen 1,000-example validation split; it does not access test predictions or train models. The human SESOI review record remains null and therefore blocks the freeze.

The 20 planned new exact trainings also have a hardware-specific planning estimate generated from the 30 condition-matched pilot runtimes: a serial median of 22.53 seconds with a 95% bootstrap interval of 22.19–22.94 seconds. This is not a fresh benchmark and excludes validation inference, startup, scheduling, memory, and transfer to any other environment; it therefore does not estimate total calibration runtime.

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